Exam 1 — Solutions

Foundations under pressure

Answer key and worked explanations. Source aliases and convention warnings are listed in `COVERAGE.md` and `SOURCE_NOTES.md` in the study pack.

Quick key

Question	Answer	Question	Answer
01	B	26	A
02	A	27	D
03	A	28	C
04	A	29	A
05	C	30	B
06	D	31	-270.00
07	C	32	0.75
08	B	33	4.00
09	C	34	1
10	A	35	8
11	C	36	8
12	B	37	2.83
13	A	38	59.04
14	D	39	2.36
15	B	40	34
16	D	41	7.33
17	C	42	48.75
18	A	43	0.63
19	D	44	0.57
20	D	45	0.33
21	C	46	1680
22	B	47	368
23	B	48	7850
24	B	49	8
25	D	50	2.00

Worked solutions

Q01 — B

B. $C = -R^{\mathsf{T}}t$
At the optical centre $X_c = 0$. Thus $RC + t = 0$ and $C = -R^{-1}t = -R^{\mathsf{T}}t$. The translation column is not generally the camera position.
Topic: camera. Sources: HC1b pp.35–45; S1.

Q02 — A

A. Halve f_x, f_y, skew, c_x and c_y; keep R and t unchanged.
Pixel coordinates change, not the camera pose. The first two rows of K scale by $1/2$ and its last row stays $\begin{bmatrix}0 & 0 & 1\end{bmatrix}$. Scaling the entire K only changes homogeneous scale.
Topic: camera. Sources: HC1b pp.35–45; S1.

Q03 — A

A. They lie on the same viewing ray from the optical centre.
Projection retains the direction of a camera-centred ray but loses distance along it. This is a fundamental single-view depth ambiguity.
Topic: camera. Sources: HC1a; HC1b pp.24–34.

Q04 — A

A. Its outgoing radiance, apart from visibility and camera effects.
Lambertian radiance is view-independent. Projected area changes with viewpoint, and brightness still depends on the light–normal angle. Do not confuse radiance with the amount of energy collected from a fixed surface area.
Topic: reflectance. Sources: HC2a pp.12–20.

Q05 — C

C. The component of g perpendicular to that plane is unobservable.
L has rank at most 2. Any vector in its nullspace can be added to g without changing Lg. More dependent observations cannot supply the missing component.
Topic: photometric. Sources: HC2a pp.34–39; Lab1.

Q06 — D

D. The normal is undefined; mask or otherwise handle the zero-reflectance/invalid pixel.
A zero vector has no direction. Adding a small denominator can prevent a numerical exception, but does not recover a physically supported normal.
Topic: photometric. Sources: Lab1; HC2a pp.35–45.

Q07 — C

C. Interior intensities increase; 0 and 1 remain fixed.
For r in $(0,1)$, reducing a positive exponent below 1 makes r^{γ} larger than r. Always state the power-law convention; some software uses its reciprocal.
Topic: histogram. Sources: HC2a pp.59–63; S1.

Q08 — B

B. Different provisional labels can later be found to belong to one connected region.
The first pass can encounter separate branches before their connecting pixels. The second pass resolves recorded label equivalences.
Topic: morphology. Sources: HC2b pp.7–12.

Q09 — C

C. Opening, then closing.
Opening removes small foreground structures. Closing fills small background gaps/holes. Either can alter genuine features, so choose element size and shape with care.
Topic: morphology. Sources: HC2b pp.13–45; S2.

Q10 — A

A. Flip K horizontally and vertically before the sliding dot product.
Convolution reverses both spatial offsets. Symmetric kernels conceal this distinction; asymmetric derivative or shift kernels do not.
Topic: filtering. Sources: HC2b pp.78–80; S2.

Q11 — C

C. 5
Variances add: $\sigma^2 = 3^2 + 4^2 = 25$. Standard deviations do not add directly.
Topic: filtering. Sources: HC2b pp.82–87; S2.

Q12 — B

B. $\sqrt{a^2 + b^2}$, $\arctan2(b, a)$
Magnitude is distance from the origin. Phase is the quadrant-aware angle measured from the positive real axis; NumPy takes (imaginary, real). See SOURCE_NOTES.md for the slide argument-order issue.
Topic: fourier. Sources: HC2b pp.97–100; SOURCE_NOTES.

Q13 — A

A. Convolve the image with the corresponding derivative of the Gaussian.
$\frac{\partial (I * G)}{\partial x} = I * \frac{\partial G}{\partial x}$. Smoothing controls noise before measuring spatial changes.
Topic: features. Sources: HC3a pp.20–54; S2; Lab2.

Q14 — D

D. Keep the connected weak chain and discard the disconnected chain.
Strong pixels seed edge following. Weak pixels survive only through a qualifying connection to a strong seed, under the declared neighbourhood rule.
Topic: features. Sources: HC3a pp.58–65; S2.

Q15 — B

B. Strong intensity variation in two independent directions: a good corner-like feature.
Both displacement directions produce a large change. One large and one small eigenvalue instead indicates an edge; two small values indicate a flat patch.
Topic: features. Sources: S3; Lab3 Sections 2–3.

Q16 — D

D. A vertical line cannot be represented by a finite m; an orthogonal-distance line model avoids this problem.
The slope-intercept model privileges y as the dependent variable. Total least squares uses perpendicular distances and a line representation that admits vertical lines.
Topic: fitting. Sources: HC3a pp.76–79; S3.

Q17 — C

C. I_t increases, often sharply.
The all-inlier sample probability is w^s. A lower w makes such samples rarer. RANSAC can work below 50% inliers, but the required budget can be large.
Topic: fitting. Sources: HC3a pp.91–109; S3.

Q18 — A

A. $A^\top A$: the local second-moment/structure tensor.
$A^\top A$ contains sums of I_x^2, $I_x I_y$ and I_y^2. Its eigenvalues determine whether the local flow estimate is well constrained in both directions.
Topic: flow. Sources: HC3b p.47; S3; Lab3.

Q19 — D

D. A single common flow vector for all pixels in the window.
A larger window adds equations but can mix incompatible motions. Local motion consistency, not simply pixel count, matters.
Topic: flow. Sources: HC3b pp.39–47; Lab3.

Q20 — D

D. Parallelism of lines.
Affine transforms preserve parallelism and ratios along a line, not general angles or lengths. General homographies preserve straight lines but can map parallel lines to intersecting ones.
Topic: geometry. Sources: HC4a pp.17–22; S4.

Q21 — C

C. One homography, independent of scene depth.
Pure rotation makes the image mapping depend on ray direction rather than depth: $H = K_2 R_{21} K_1^{-1}$. A planar scene is the other important homography case.
Topic: geometry. Sources: HC4a pp.33–35; S4.

Q22 — B

B. Without a scale constraint, $h = 0$ is a trivial minimizer and does not define a homography.
A homography is defined only up to nonzero scale. The constrained solution is the smallest-eigenvalue eigenvector of $A^\top A$, equivalently the appropriate right singular vector of A.
Topic: geometry. Sources: HC4a pp.51–55.

Q23 — B

B. The match is ambiguous even if the nearest absolute distance is small.
Repeated or weakly distinctive patterns can have several nearly equally plausible matches. The ratio compares distinctiveness relative to the next alternative.
Topic: matching. Sources: HC4a pp.41–44; S4; Lab4.

Q24 — B

B. Represent them identically, because it discards spatial arrangement.
The vocabulary gives fixed-length count vectors. Plain BoVW is orderless; a spatial extension would be needed to retain layout.
Topic: classification. Sources: HC4b pp.23–28; S4.

Q25 — D

D. Fit the vocabulary on training descriptors; select settings on validation data; evaluate once on untouched test data.
Unsupervised preprocessing can still leak information. The validation set selects hyperparameters; the test set measures the final chosen pipeline.
Topic: classification. Sources: HC4b pp.25–28,55–58; Lab5.

Q26 — A

A. 0
I_t finds none of the minority examples: TP $= 0$ and FN is the minority count. Accuracy alone can hide complete failure on a rare class.
Topic: metrics. Sources: HC4b pp.80–86.

Q27 — D

D. Translation equivariance: shifting the input shifts the feature map.
Shared local operations follow translations. Invariance would mean unchanged output; pooling and aggregation can provide limited or additional invariance, not the same property.
Topic: cnn. Sources: HC5b pp.22–24,64; S5.

Q28 — C

C. Faster R-CNN uses a learned Region Proposal Network; Fast R-CNN normally relies on external proposals such as Selective Search.
Fast R-CNN shares a full-image convolutional feature map but retains external proposals. Faster R-CNN also learns proposals through an RPN sharing backbone features.
Topic: detection. Sources: HC5b pp.125–133; HC6a pp.18–38.

Q29 — A

A. ResNet adds compatible activations; U-Net concatenates encoder and decoder features across matching scales.
Residual addition supports identity-like mappings and gradient paths. U-Net concatenation transfers high-resolution spatial information to the decoder.
Topic: architectures. Sources: HC6a pp.99–112,122–137.

Q30 — B

B. An epipolar line on which the corresponding point must lie.
Epipolar geometry reduces correspondence search from 2D to 1D. I_t does not select a unique point along the line. The nonzero condition excludes the epipole's degenerate zero output.
Topic: stereo. Sources: HC6b pp.50–65.

Q31 — -270.00

Submit -270.00. Unrounded reference: `-270.0`.
$X_w - C = \begin{bmatrix}2 & 3 & 4\end{bmatrix}^\mathsf{T}$, $X_c = \begin{bmatrix}-3 & 2 & 4\end{bmatrix}^\mathsf{T}$. $q_0 = 800(-3) + 20 \cdot 2 + 320 \cdot 4 = -1080$. Divide by 4: $u = -270$. A negative pixel coordinate can simply be outside the field of view.
Topic: camera. Sources: HC1b pp.24–45; S1.

Q32 — 0.75

Submit 0.75. Unrounded reference: `0.7452154112317649`.
$L^\mathsf{T}L = \mathrm{diag}(1.28, 1.28, 2.44)$ and $L^\mathsf{T}i = \begin{bmatrix}0.384 & 0.064 & 1.66\end{bmatrix}^\mathsf{T}$. Thus $g = \begin{bmatrix}0.3 & 0.05 & 1.66/2.44\end{bmatrix}^\mathsf{T}$. Its norm is about 0.7452, so submit 0.75. Do not normalize g before taking its magnitude.
Topic: photometric. Sources: HC2a pp.34–39; Lab1; Coding examples Q1,Q5.

Q33 — 4.00

Submit 4.00. Unrounded reference: `4.0`.
$p = -\frac{2}{2} = -1$ and $q = -\frac{-1}{2} = 0.5$. $z(3,4) = 5 + 3(-1) + 4 \cdot 0.5 = 4$. The declared sign convention is essential.
Topic: photometric. Sources: HC2a pp.41–43; Lab1; SOURCE_NOTES.

Q34 — 1

Submit 1. Unrounded reference: `1`.
$\mathrm{CDF}(1) = \frac{1+3}{8} = 0.5$; $\lfloor 3 \cdot 0.5 \rfloor = 1$. Other equalization conventions can differ, which is why the mapping is specified.
Topic: histogram. Sources: HC2a pp.51–63; S1.

Q35 — 8

Submit 8. Unrounded reference: `8`.
Dilation gives $\begin{bmatrix}1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0\end{bmatrix}$. Erosion of that fixed-length result gives $\begin{bmatrix}0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0\end{bmatrix}$, which has 8 foreground entries.
Topic: morphology. Sources: HC2b pp.13–45; S2.

Q36 — 8

Submit 8. Unrounded reference: `8`.
The flipped kernel is

$$\begin{bmatrix}-2 & 0 & 3\\ 0 & -1 & 0\\ 2 & 0 & 1\end{bmatrix}.$$

The dot product is $-2 + 1 + 4 + 5 = 8$. The unflipped result would be different.
Topic: filtering. Sources: HC2b pp.78–80; S2; Lab2.

Q37 — 2.83

Submit 2.83. Unrounded reference: `2.8284271247461903`.
$F_1 = 1 - 2\mathrm{i} - 3 + 4\mathrm{i} = -2 + 2\mathrm{i}$. Its magnitude is $\sqrt{4+4} = \sqrt{8}$, so submit 2.83.
Topic: fourier. Sources: HC2b pp.95–104.

Q38 — 59.04

Submit 59.04. Unrounded reference: `59.04`.
$\det(M) = 81 - 9 = 72$; $\mathrm{tr}(M) = 18$. $R = 72 - 0.04 \cdot 324 = 59.04$. A positive response alone still needs thresholding and spatial non-maximum suppression.
Topic: features. Sources: S3; Lab3.

Q39 — 2.36

Submit 2.36. Unrounded reference: `2.3636363636363638`.
$A^{\mathsf{T}}A =$

$$\begin{bmatrix}6 & 1\\ 1 & 2\end{bmatrix},$$

$A^{\mathsf{T}}(-I_t) = \begin{bmatrix}13 & 0\end{bmatrix}^{\mathsf{T}}$. Solve $6u + v = 13$ and $u + 2v = 0$: $u = \frac{26}{11}$, $v = -\frac{13}{11}$. Submit 2.36; forgetting the minus sign reverses the estimate.
Topic: flow. Sources: HC3b pp.39–47; S3; Lab3.

Q40 — 34

Submit 34. Unrounded reference: `34`.
The all-inlier probability is $0.6^4 = 0.1296$. The ratio of logarithms is approximately 33.18; round upwards, not to the nearest integer. $N = 34$.
Topic: fitting. Sources: HC3a pp.104–105; HC4a pp.45–48.

Q41 — 7.33

Submit 7.33. Unrounded reference: `7.333333333333333`.
$H\begin{bmatrix}5 & 4 & 1\end{bmatrix}^\top = \begin{bmatrix}11 & 2 & 1.5\end{bmatrix}^\top$. Dehomogenize: $x' = \frac{11}{1.5} = 7.3333$, so submit 7.33. The third coordinate is not generally 1.
Topic: geometry. Sources: HC4a pp.20–22,51; S4.

Q42 — 48.75

Submit 48.75. Unrounded reference: `48.75`.
Weights are 0.1875, 0.0625, 0.5625 and 0.1875. The weighted sum is $1.875 + 1.875 + 28.125 + 16.875 = 48.75$. Do not interchange x and y.
Topic: geometry. Sources: HC4a pp.24–31; Lab4.

Q43 — 0.63

Submit 0.63. Unrounded reference: `0.631578947368421`.
$\mathrm{TP} = 12,\ \mathrm{FP} = 2 + 4 = 6,\ \mathrm{FN} = 3 + 5 = 8$. $F_1 = \frac{2\mathrm{TP}}{2\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} = \frac{24}{38} = 0.6315789$. Submit 0.63.
Topic: metrics. Sources: HC4b pp.80–86; S4.

Q44 — 0.57

Submit 0.57. Unrounded reference: `0.5666666666666667`.
$\mathrm{AP} = \frac{1/1+2/3+3/5}{4} = \frac{17}{30}$. The missing relevant image contributes zero, so the denominator is 4, not 3. Submit 0.57.
Topic: metrics. Sources: HC5a pp.19–24,68–78; S4.

Q45 — 0.33

Submit 0.33. Unrounded reference: `0.3333333333333333`.
Each area is 24. The intersection is width 4 times height $3 = 12$. Union $= 24 + 24 - 12 = 36$. IoU $= \frac{12}{36} = \frac{1}{3}$; submit 0.33.
Topic: detection. Sources: HC5a p.62; S4.

Q46 — 1680

Submit 1680. Unrounded reference: `1680`.
$H_{\text{out}} = \left\lfloor \frac{31+2-5}{2} \right\rfloor + 1 = 15$; $W_{\text{out}} = \left\lfloor \frac{29-3}{2} \right\rfloor + 1 = 14$. Total $= 15 \cdot 14 \cdot 8 = 1680$. Count output positions, not input positions.
Topic: cnn. Sources: HC5b pp.25–43,66–68; S5.

Q47 — 368

Submit 368. Unrounded reference: `368`.
Each output channel has $5 \cdot 3 \cdot 3 = 45$ weights and one bias. There are 8 output channels, so $8(45+1) = 368$. Spatial resolution and stride do not change this count.
Topic: cnn. Sources: HC5b pp.66–73; S5.

Q48 — 7850

Submit 7850. Unrounded reference: `7850`.
The flattened input has $7 \cdot 7 \cdot 16 = 784$ values. Count $784 \cdot 10$ weights plus 10 biases $= 7850$.
Topic: cnn. Sources: HC5b pp.16–24; S5.

Q49 — 8

Submit 8. Unrounded reference: `8`.
After conv: $r = 3$, $j = 1$. After pool: $r = 4$, $j = 2$. After second conv: $r = 4 + 2 \cdot 2 = 8$, $j = 2$. Pooling changes the spacing used by later layers.
Topic: cnn. Sources: HC5b pp.80–85; HC6a pp.6–9.

Q50 — 2.00

Submit 2.00. Unrounded reference: `2.0`.
$l_2 = \begin{bmatrix} 0 & -1 & 7 \end{bmatrix}^\intercal$, so the line is $y = 7$. The candidate has $y = 9$, giving distance $\frac{|-9+7|}{1} = 2$. This tests the line constraint, not whether the candidate is the true appearance match.
Topic: stereo. Sources: HC6b pp.50–65.